

European Sovereign Green Bond Event Study

Thesis: When Governments Say Green - Do Bond Markets Believe Them? **Author:** Nahian Ibnat | **Supervisor:** Zoltan Csaba Toth **Institution:** Central European University | **Date:** May 2026

Role of this notebook

This notebook implements the event study for Germany and Denmark - the comparative European extension. Asia produced null H3 results. Europe tests whether the same methodology detects greeniums in deeper, more liquid markets.

Critical economic corrections from the prior draft

Issue	What was wrong	Fix
Germany pre-event weeks misinterpreted	Described as parallel trends test, but green bond did not exist before treatment date	Reframed as post-issuance price discovery, not parallel trends
Denmark same problem	Conventional twin exists pre-issuance but green bond does not - no pre-treatment greenium	Acknowledged; parallel trends for Denmark comes from stacked DiD Stack A
Pooled event study limitation undocumented	Germany and Denmark combined with no warning about conflation of different macro periods	Explicit warning added
Summary false positive	Any significant week counted as treatment detected	Fixed
No formal test	Pre-trend assessed visually	Joint F-test on post-event dummies added
FX/CDS not forward-filled	Weekend gaps silently dropped	Added

Critical economic insight - why Europe is different from Asia

In Asia, India has a genuine pre-treatment window of ~120 trading days because the second auction treatment event (Oct 2023) occurs months after the green bond started trading (Jan

2023).

In Europe, BOTH Germany and Denmark issued green bonds FOR THE FIRST TIME at the treatment date. The greenium cannot exist before the green bond exists. Therefore:

- There is NO genuine pre-treatment greenium data for Germany or Denmark
- The event study documents POST-ISSUANCE dynamics, not a classic DiD
- Parallel trends for Europe is validated by the stacked DiD (regression notebook)
- The event study here shows how quickly and persistently the greenium established

Country	Pre-treatment greenium data	Parallel trends testable here
India	YES (~120 trading days)	YES
Indonesia	Very limited (~55 days)	Weakly
Germany	NO (bond issued at treatment)	NO
Denmark	NO (bond issued at treatment)	NO

Structure

Part	Content
1	Setup
2	Data loading
3	Germany - post-issuance price discovery
4	Denmark - post-issuance dynamics
5	Pooled event study (with limitation note)
6	Side-by-side comparison
7	Joint F-test on post-event dummies
8	Summary and thesis implications

Part 1 - Setup

1.1 - Install dependencies

```
In [1]: !pip install -q openpyxl pandas numpy statsmodels matplotlib scipy
print("OK All packages installed.")
```

OK All packages installed.

1.2 - Imports

```
In [2]: import os
import warnings
import numpy as np
import pandas as pd
import statsmodels.api as sm
from statsmodels.regression.linear_model import OLS
from statsmodels.stats.stattools import durbin_watson
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
import openpyxl
warnings.filterwarnings("ignore")
print("OK Libraries imported.")
```

OK Libraries imported.

1.3 - File paths

```
In [3]: ROOT = ".."

BOND_FILES = {
    "DEU_GREEN_10Y": os.path.join(ROOT, "data", "raw", "bond_yield_data", "germany_
        "DEU_GREEN_10Y_DE0001030708.xlsx"),
    "DEU_CONV_10Y": os.path.join(ROOT, "data", "raw", "bond_yield_data", "germany_
        "DEU_CONV_10Y_DE0001102507.xlsx"),
    "DNK_GREEN_10Y": os.path.join(ROOT, "data", "raw", "bond_yield_data", "denmark_
        "DNK_GREEN_10Y_DK0009924375.xlsx"),
    "DNK_CONV_10Y": os.path.join(ROOT, "data", "raw", "bond_yield_data", "denmark_
        "DNK_CONV_10Y_DK0009924102.xlsx"),
}

VIX_FILE = os.path.join(ROOT, "data", "raw", "control_variables", "vix", "VIX
EURUSD_FILE = os.path.join(ROOT, "data", "raw", "control_variables", "exchange_r
    "CTRL_EURUSD_01Jan2019_18May2026.csv")
USDDKK_FILE = os.path.join(ROOT, "data", "raw", "control_variables", "exchange_r
    "CTRL_USDDKK_01Jan2019_18May2026.csv")
CDS_DEU_FILE = os.path.join(ROOT, "data", "raw", "control_variables", "cds",
    "CTRL_CDS5Y_Germany_DEGV5YUSAC.csv")
SENTIMENT_FILE = os.path.join(ROOT, "data", "processed", "gdelt",
    "daily_sentiment_gdelt_europe.csv")
OUTPUT_FOLDER = os.path.join(ROOT, "output", "regression_europe")
os.makedirs(OUTPUT_FOLDER, exist_ok=True)

PAIRS = [
    ("DEU_GREEN_10Y", "DEU_CONV_10Y", "Germany", "10Y"),
    ("DNK_GREEN_10Y", "DNK_CONV_10Y", "Denmark", "10Y"),
]

TREATMENT_DATES = {
    "Germany": pd.Timestamp("2020-09-02"),
    "Denmark": pd.Timestamp("2022-01-21"),
}
```

```

print("Path verification:")
for name, path in [("DEU_GREEN_10Y", BOND_FILES["DEU_GREEN_10Y"]),
                  ("DEU_CONV_10Y", BOND_FILES["DEU_CONV_10Y"]),
                  ("DNK_GREEN_10Y", BOND_FILES["DNK_GREEN_10Y"]),
                  ("DNK_CONV_10Y", BOND_FILES["DNK_CONV_10Y"]),
                  ("VIX", VIX_FILE),
                  ("Sentiment", SENTIMENT_FILE)]:
    status = "OK" if os.path.exists(path) else "NOT FOUND"
    print(f" {name:<22} {status}")
print()
for c, d in TREATMENT_DATES.items():
    print(f" Treatment {c:<10} {d.date()}")
print("\nSetup complete.")

```

Path verification:

DEU_GREEN_10Y	OK
DEU_CONV_10Y	OK
DNK_GREEN_10Y	OK
DNK_CONV_10Y	OK
VIX	OK
Sentiment	OK

Treatment Germany	2020-09-02
Treatment Denmark	2022-01-21

Setup complete.

1.4 - Helper functions and chart styling

```

In [4]: def load_bond_ytm(filepath):
        wb = openpyxl.load_workbook(filepath)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        hdr_idx = next(i for i, r in enumerate(rows) if r[0] and "date" in str(r[0]).lower())
        df = pd.DataFrame([r for r in rows[hdr_idx+1:] if r[0] is not None], columns=rows[hdr_idx][0])
        df["date"] = pd.to_datetime(df["Date"])
        df["ytm"] = (df["BidYld"] + df["AskYld"]) / 2
        return df[["date", "ytm"]].dropna().sort_values("date").reset_index(drop=True)

def load_investing_fx(filepath, col_name):
    df = pd.read_csv(filepath)
    df["date"] = pd.to_datetime(df["Date"])
    df[col_name] = pd.to_numeric(df["Price"].astype(str).str.replace(",", ""))
    return df[["date", col_name]].dropna().sort_values("date").reset_index(drop=True)

def load_cds(filepath, col_name):
    df = pd.read_csv(filepath)
    df.columns = ["date", col_name, "open", "high", "low", "chg"]
    df["date"] = pd.to_datetime(df["date"])
    df[col_name] = pd.to_numeric(df[col_name], errors="coerce")
    return df[["date", col_name]].dropna().sort_values("date").reset_index(drop=True)

BLUE="#378ADD"; ORANGE="#E85D24"; TEAL="#1D9E75"
GRAY="#888888"; WHITE="#f2f2f2"; BLACK="#222222"; BG="white"; PANEL_BG="white"

```



```
def style_ax(ax):
    ax.set_facecolor('white')
    ax.tick_params(colors='#222222', labelsiz=8)
    ax.xaxis.label.set_color('#222222'); ax.yaxis.label.set_color('#222222')
    ax.title.set_color('#222222')
    for s in ax.spines.values(): s.set_edgecolor('#cccccc')
    ax.spines["top"].set_visible(False); ax.spines["right"].set_visible(False)

print("OK Helper functions defined.")
```

OK Helper functions defined.

Part 2 - Data Loading and Panel Construction

2.1 - Bond yields and greenium

Both European pairs are exact twins: identical coupon (0%), identical maturity date, same sovereign issuer. The greenium is a model-free pure green label premium.

```
In [5]: bond_dfs = {}
for key, fpath in BOND_FILES.items():
    bond_dfs[key] = load_bond_ytm(fpath)

panels = []
for gkey, ckey, country, tenor in PAIRS:
    g = bond_dfs[gkey].rename(columns={"ytm": "ytm_green"})
    c = bond_dfs[ckey].rename(columns={"ytm": "ytm_conv"})
    m = pd.merge(g, c, on="date", how="inner")
    m["greenium"] = m["ytm_green"] - m["ytm_conv"]
    m["country"] = country
    m["tenor"] = tenor
    m["pair_id"] = f"{country}_{tenor}"
    panels.append(m)

panel = pd.concat(panels, ignore_index=True)
print(f"Raw panel: {len(panel):,} obs")
print(panel.groupby(["country", "tenor"]).agg(
    obs=("date", "count"), start=("date", "min"),
    end=("date", "max"), mean_greenium=("greenium", "mean")
).round(4).to_string())
```

Raw panel: 2,620 obs

		obs	start	end	mean_greenium
country	tenor				
Denmark	10Y	1132	2022-01-14	2026-05-18	-0.0278
Germany	10Y	1488	2020-09-03	2026-05-18	-0.0182

2.2 - Merge controls

```
In [6]: vix = pd.read_csv(VIX_FILE)
vix.columns = ["date", "vix"]
vix["date"] = pd.to_datetime(vix["date"])
vix = vix[vix["vix"] != "."].copy()
vix["vix"] = pd.to_numeric(vix["vix"])

eurusd = load_investing_fx(EURUSD_FILE, "eurusd")
usddkk = load_investing_fx(USDDKK_FILE, "usddkk")
usddkk["dkkusd"] = 1 / usddkk["usddkk"]
usddkk = usddkk[["date", "dkkusd"]]
cds_deu = load_cds(CDS_DEU_FILE, "cds_germany")

sent = pd.read_csv(SENTIMENT_FILE, parse_dates=["date"])
sent_deu = sent[sent["country"]=="Germany"][["date", "sentiment_gdelt", "sentiment_st"]
sent_deu.columns = ["date", "sentiment", "signaling_noise"]
sent_dnk = sent[sent["country"]=="Denmark"][["date", "sentiment_gdelt", "sentiment_st"]
sent_dnk.columns = ["date", "sentiment", "signaling_noise"]

deu_p = panel[panel["country"]=="Germany"].merge(sent_deu, on="date", how="left")
dnk_p = panel[panel["country"]=="Denmark"].merge(sent_dnk, on="date", how="left")
panel = pd.concat([deu_p, dnk_p], ignore_index=True)

vix_ff = (vix.set_index("date")
          .reindex(pd.date_range(panel.date.min(), panel.date.max()))
          .ffill().reset_index())
vix_ff.columns = ["date", "vix"]
panel = panel.merge(vix_ff, on="date", how="left")

panel = panel.merge(eurusd, on="date", how="left")
panel = panel.merge(usddkk, on="date", how="left")
panel["fx"] = np.where(panel["country"]=="Germany", panel["eurusd"], panel["dkkusd"])
panel = panel.drop(columns=["eurusd", "dkkusd"])

panel = panel.merge(cds_deu, on="date", how="left")
panel["cds"] = panel["cds_germany"]
panel = panel.drop(columns=["cds_germany"])

# Forward-fill FX and CDS gaps within country
panel = panel.sort_values(["pair_id", "date"]).reset_index(drop=True)
panel["fx"] = panel.groupby("country")["fx"].ffill()
panel["cds"] = panel.groupby("country")["cds"].ffill()

panel["log_fx"] = np.log(panel["fx"].clip(lower=1e-6))
panel["log_vix"] = np.log(panel["vix"].clip(lower=1e-6))

panel = panel.dropna(subset=["sentiment", "signaling_noise", "vix", "cds", "fx"]).reset
print(f"Panel after merge: {len(panel):,} obs")
print(panel.groupby(["country", "tenor"]).agg(obs=("date", "count"), start=("date", "m
print("OK Controls merged. FX and CDS forward-filled.")
```

Panel after merge: 2,574 obs

	obs	start	end
country tenor			
Denmark 10Y	1109	2022-01-14	2026-05-01
Germany 10Y	1465	2020-09-03	2026-05-01

OK Controls merged. FX and CDS forward-filled.

2.3 - First-differencing

```
In [7]: panel = panel.sort_values(["pair_id", "date"]).reset_index(drop=True)

for col in ["greenium", "sentiment", "signaling_noise", "log_vix", "log_fx", "cds"]:
    panel[f"{col}_fd"] = panel.groupby("pair_id")[col].diff()

panel_fd = panel.dropna(subset=["greenium_fd"]).reset_index(drop=True)

for col in ["sentiment_fd", "signaling_noise_fd", "log_vix_fd", "log_fx_fd", "cds_fd"]:
    mu = panel_fd[col].mean(); sd = panel_fd[col].std()
    panel_fd[f"{col}_z"] = (panel_fd[col] - mu) / sd if sd > 0 else 0.0

print(f"First-differenced panel: {len(panel_fd):,} obs")
print("NaN check:")
for col in ["greenium_fd", "sentiment_fd_z", "log_vix_fd_z", "cds_fd_z", "log_fx_fd_z"]:
    print(f" {col:<30} NaN={panel_fd[col].isna().sum()}")
print("OK Ready for event study.")
```

First-differenced panel: 2,572 obs

NaN check:

greenium_fd	NaN=0
sentiment_fd_z	NaN=0
log_vix_fd_z	NaN=0
cds_fd_z	NaN=0
log_fx_fd_z	NaN=0

OK Ready for event study.

Part 3 - Germany: Post-Issuance Price Discovery

Economic framing

Germany has NO pre-treatment greenium data. The green Bund (DE0001030708) was issued on 2 Sep 2020. Before that date the bond did not exist. The event study for Germany therefore does NOT test parallel trends in the DiD sense.

What it shows instead: the price discovery dynamic in weeks after inaugural issuance. Did the greenium establish itself immediately, or did it take weeks to settle? This complements the regression results by documenting the temporal pattern, not providing causal identification.

The DiD identification for Germany relies on the twin-bond design (identical coupon 0%, identical maturity Aug 2030) and issuance date exogeneity.

NOTE on pre-event weeks: Week -1 is the week immediately before issuance. Since the green bond started trading AT the issuance date, pre-event weeks have very few or zero observations and should NOT be interpreted as a pre-treatment baseline.

3.1 - Assign event time (Germany)

```
In [8]: panel_deu_es = panel_fd[panel_fd["country"]=="Germany"].copy().reset_index(drop=True)

treat_date_deu = TREATMENT_DATES["Germany"]
country_dates = np.sort(panel_deu_es["date"].unique())
treat_idx      = np.searchsorted(country_dates, np.datetime64(treat_date_deu))
date_to_event  = {d: int(i - treat_idx) for i, d in enumerate(country_dates)}
panel_deu_es["event_time"] = panel_deu_es["date"].map(date_to_event)
panel_deu_es["event_week"] = (panel_deu_es["event_time"] / 5).apply(np.floor).astype(int)

WEEK_MIN_DEU, WEEK_MAX_DEU = -5, 20
panel_deu_es = panel_deu_es[panel_deu_es["event_week"].between(WEEK_MIN_DEU, WEEK_MAX_DEU)]

print(f"Germany: {len(panel_deu_es):,} obs | {treat_date_deu.date()} | weeks {WEEK_MIN_DEU} to {WEEK_MAX_DEU}")
print("NOTE: Pre-event weeks (negative) have very few obs - bond did not exist before issuance")
for w, n in panel_deu_es.groupby("event_week").size().items():
    note = " <- THIN: bond barely existed" if w < 0 else " <- issuance week" if w == 0 else ""
    print(f"  Week {w:>4}: {n:>3} obs{note}")
```

Germany: 105 obs | 2020-09-02 | weeks -5 to 20

NOTE: Pre-event weeks (negative) have very few obs - bond did not exist before issuance

Week	0:	5 obs <- issuance week
Week	1:	5 obs
Week	2:	5 obs
Week	3:	5 obs
Week	4:	5 obs
Week	5:	5 obs
Week	6:	5 obs
Week	7:	5 obs
Week	8:	5 obs
Week	9:	5 obs
Week	10:	5 obs
Week	11:	5 obs
Week	12:	5 obs
Week	13:	5 obs
Week	14:	5 obs
Week	15:	5 obs
Week	16:	5 obs
Week	17:	5 obs
Week	18:	5 obs
Week	19:	5 obs
Week	20:	5 obs

3.2 - Germany regression

```
In [9]: ev_dummies_deu = pd.get_dummies(panel_deu_es["event_week"].astype(str),
                                         drop_first=False, dtype=float)

if "-1" in ev_dummies_deu.columns:
    ev_dummies_deu = ev_dummies_deu.drop(columns=["-1"])

panel_deu_reg = pd.concat([panel_deu_es, ev_dummies_deu], axis=1)
ev_cols_deu   = list(ev_dummies_deu.columns)

X_deu        = sm.add_constant(panel_deu_reg[ev_cols_deu])
model_deu    = OLS(panel_deu_reg["greenium_fd"], X_deu).fit(cov_type="HC3")
all_weeks_deu = sorted(range(WEEK_MIN_DEU, WEEK_MAX_DEU + 1))
ci_deu       = model_deu.conf_int(alpha=0.05)
coefs_deu, lo_deu, hi_deu, p_deu = [], [], [], []

for w in all_weeks_deu:
    col = str(w)
    if col in model_deu.params.index:
        coefs_deu.append(model_deu.params[col])
        lo_deu.append(ci_deu.loc[col, 0])
        hi_deu.append(ci_deu.loc[col, 1])
        p_deu.append(model_deu.pvalues[col])
    else:
        coefs_deu.append(0.0); lo_deu.append(0.0); hi_deu.append(0.0); p_deu.append(0.0)

print(f"Germany: N={int(model_deu.nobs):,} R2={model_deu.rsquared:.4f} DW={durbin
print("\nGERMANY POST-ISSUANCE PRICE DISCOVERY (not parallel trends)")
print("=" * 70)
print(f"{'Week':<8} {'Period':<10} {'Coef':>10} {'CI Lo':>10} {'CI Hi':>10} {'p':>8}
print("-" * 70)
for w, c, lo, hi, p in zip(all_weeks_deu, coefs_deu, lo_deu, hi_deu, p_deu):
    period = "Pre(*)" if w < -1 else "Ref  " if w == -1 else "Post  "
    ref    = "(ref)" if w == -1 else ("****" if p<0.01 else "****" if p<0.05 else "")
    print(f"{'w':<8} {'period':<10} {'c':>+10.4f} {'lo':>10.4f} {'hi':>10.4f} {'p':>8.4f} {ref}")
print("\n(*) Pre-event weeks are economically thin. Do NOT interpret as parallel tr
```

Germany: N=105 R2=0.0827 DW=2.8454

GERMANY POST-ISSUANCE PRICE DISCOVERY (not parallel trends)

Week	Period	Coef	CI Lo	CI Hi	p Sig
-5	Pre(*)	+0.0000	0.0000	0.0000	1.0000
-4	Pre(*)	+0.0000	0.0000	0.0000	1.0000
-3	Pre(*)	+0.0000	0.0000	0.0000	1.0000
-2	Pre(*)	+0.0000	0.0000	0.0000	1.0000
-1	Ref	+0.0000	0.0000	0.0000	1.0000 (ref)
0	Post	-0.0021	-0.0059	0.0017	0.2784
1	Post	-0.0004	-0.0047	0.0039	0.8585
2	Post	-0.0006	-0.0031	0.0019	0.6457
3	Post	+0.0007	-0.0014	0.0027	0.5280
4	Post	-0.0010	-0.0043	0.0023	0.5613
5	Post	+0.0008	-0.0005	0.0022	0.2341
6	Post	+0.0002	-0.0064	0.0069	0.9435
7	Post	-0.0008	-0.0023	0.0008	0.3156
8	Post	+0.0000	-0.0024	0.0025	0.9717
9	Post	+0.0033	-0.0018	0.0085	0.1997
10	Post	-0.0004	-0.0024	0.0017	0.7306
11	Post	+0.0000	-0.0013	0.0013	0.9580
12	Post	+0.0002	-0.0022	0.0026	0.8607
13	Post	+0.0000	-0.0012	0.0013	0.9392
14	Post	-0.0010	-0.0081	0.0061	0.7859
15	Post	+0.0012	0.0000	0.0024	0.0464 **
16	Post	+0.0000	-0.0043	0.0043	0.9915
17	Post	-0.0012	-0.0085	0.0062	0.7544
18	Post	+0.0002	-0.0012	0.0016	0.7367
19	Post	+0.0000	-0.0022	0.0022	0.9785
20	Post	+0.0002	-0.0033	0.0037	0.9032

(*) Pre-event weeks are economically thin. Do NOT interpret as parallel trends.

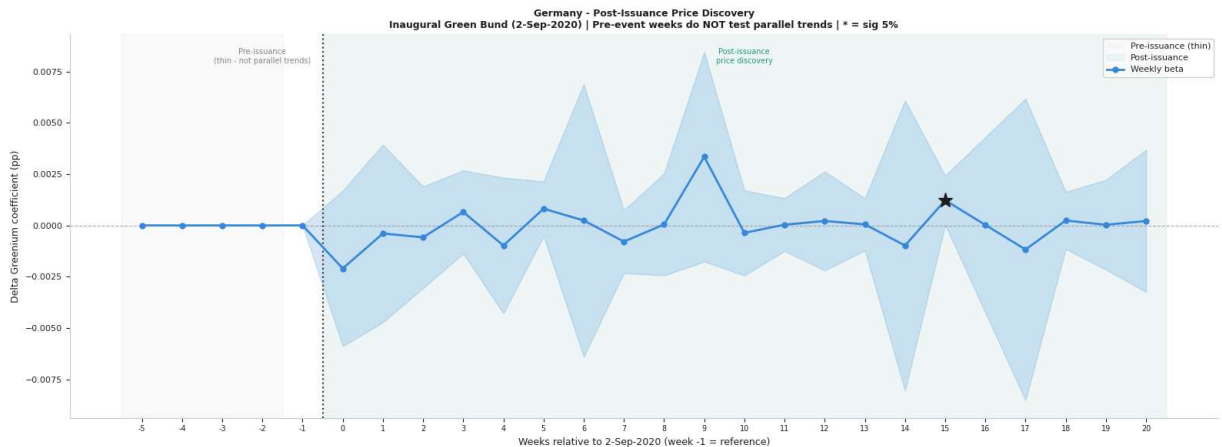
3.3 - Germany chart

```
In [10]: fig, ax = plt.subplots(figsize=(16, 6))
fig.patch.set_facecolor('white'); style_ax(ax)
x_d=np.array(all_weeks_deu); c_d=np.array(coefs_deu); lo_d=np.array(lo_deu); hi_d=n
ax.axvspan(x_d.min()-0.5,-1.5,alpha=0.03,color=GRAY,label="Pre-issuance (thin)")
ax.axvspan(-0.5,x_d.max()+0.5,alpha=0.06,color=TEAL,label="Post-issuance")
ax.fill_between(x_d,lo_d,hi_d,alpha=0.2,color=BLUE)
ax.plot(x_d,c_d,color=BLUE,linewidth=2.0,marker="o",markersize=5,label="Weekly beta
for w,c,p in zip(all_weeks_deu,coefs_deu,p_deu):
    if p<0.05 and w!=-1: ax.plot(w,c,marker="**",color=BLACK,markersize=14,zorder=4)
ax.axhline(0,color=GRAY,linewidth=0.8,linestyle="--",alpha=0.7)
ax.axvline(-0.5,color=BLACK,linewidth=1.5,linestyle=":",alpha=0.9)
ylim=ax.get_ylim()
ax.text(-2, ylim[1]*0.85, "Pre-issuance\n(thin - not parallel trends)",
        fontsize=7, color=GRAY, ha="center")
ax.text(10, ylim[1]*0.85, "Post-issuance\nprice discovery",
        fontsize=7, color=TEAL, ha="center")
ax.set_xlabel("Weeks relative to 2-Sep-2020 (week -1 = reference)",fontsize=9)
```

```

ax.set_ylabel("Delta Greenium coefficient (pp)", fontsize=9)
ax.set_title("Germany - Post-Issuance Price Discovery\n"
            "Inaugural Green Bund (2-Sep-2020) | Pre-event weeks do NOT test paral
            fontsize=9, fontweight="bold", color=BLACK)
ax.set_xticks(x_d); ax.set_xticklabels([str(w) for w in all_weeks_deu], fontsize=7, c
ax.legend(fontsize=8, facecolor='white', edgecolor="#cccccc", labelcolor=BLACK)
plt.tight_layout()
out=os.path.join(OUTPUT_FOLDER, "event_study_germany.png")
plt.savefig(out, dpi=150, bbox_inches="tight", facecolor='white'); plt.show()
print(f"Saved: {out}")

```



Saved: ..\output\regression_europe\event_study_germany.png

Part 4 - Denmark: Post-Issuance Dynamics

Economic framing

Denmark issued its inaugural green bond on 21 Jan 2022. Like Germany, the green bond did not exist before the issuance date. The pre-event weeks in the event window do not represent a genuine pre-treatment greenium baseline.

The parallel trends assumption for Denmark is validated through Stack A of the stacked DiD in the regression notebook, where Germany (already treated Sep 2020) serves as the control for Denmark issuance in Jan 2022. That is the proper identification validation for Denmark, not this event study.

What this event study shows: how Denmark greenium evolved in the weeks after inaugural issuance. Given that Denmark is described as showing a textbook greenium compression pattern (sharp sustained drop at issuance), the post-event coefficient should show a significant negative level shift.

4.1 - Assign event time (Denmark)


```
In [11]: panel_dnk_es = panel_fd[panel_fd["country"]=="Denmark"].copy().reset_index(drop=True)

treat_date_dnk = TREATMENT_DATES["Denmark"]
country_dates = np.sort(panel_dnk_es["date"].unique())
treat_idx = np.searchsorted(country_dates, np.datetime64(treat_date_dnk))
date_to_event = {d: int(i - treat_idx) for i, d in enumerate(country_dates)}
panel_dnk_es["event_time"] = panel_dnk_es["date"].map(date_to_event)
panel_dnk_es["event_week"] = (panel_dnk_es["event_time"] / 5).apply(np.floor).astype(int)

WEEK_MIN_DNK, WEEK_MAX_DNK = -10, 20
panel_dnk_es = panel_dnk_es[panel_dnk_es["event_week"].between(WEEK_MIN_DNK, WEEK_MAX_DNK)]

print(f"Denmark: {len(panel_dnk_es):,} obs | {treat_date_dnk.date()} | weeks {WEEK_MIN_DNK} to {WEEK_MAX_DNK}")
pre_obs = (panel_dnk_es["event_week"] < 0).sum()
print(f"Pre-event obs: {pre_obs} (NOTE: bond did not exist before issuance - thin data)")
for w, n in panel_dnk_es.groupby("event_week").size().items():
    marker = " <- issuance week" if w==0 else " <- reference (omitted)" if w==-1 else ""
    print(f" Week {w:>4}: {n:>3} obs{marker}")
```

Denmark: 109 obs | 2022-01-21 | weeks -10 to 20

Pre-event obs: 4 (NOTE: bond did not exist before issuance - thin data)

```
Week -1: 4 obs <- reference (omitted)
Week 0: 5 obs <- issuance week
Week 1: 5 obs
Week 2: 5 obs
Week 3: 5 obs
Week 4: 5 obs
Week 5: 5 obs
Week 6: 5 obs
Week 7: 5 obs
Week 8: 5 obs
Week 9: 5 obs
Week 10: 5 obs
Week 11: 5 obs
Week 12: 5 obs
Week 13: 5 obs
Week 14: 5 obs
Week 15: 5 obs
Week 16: 5 obs
Week 17: 5 obs
Week 18: 5 obs
Week 19: 5 obs
Week 20: 5 obs
```

4.2 - Denmark regression

```
In [12]: ev_dummies_dnk = pd.get_dummies(panel_dnk_es["event_week"].astype(str),
                                         drop_first=False, dtype=float)

if "-1" in ev_dummies_dnk.columns:
    ev_dummies_dnk = ev_dummies_dnk.drop(columns=["-1"])

panel_dnk_reg = pd.concat([panel_dnk_es, ev_dummies_dnk], axis=1)
ev_cols_dnk = list(ev_dummies_dnk.columns)
```



```

X_dnk      = sm.add_constant(panel_dnk_reg[ev_cols_dnk])
model_dnk  = OLS(panel_dnk_reg["greenium_fd"], X_dnk).fit(cov_type="HC3")
all_weeks_dnk = sorted(range(WEEK_MIN_DNK, WEEK_MAX_DNK + 1))
ci_dnk     = model_dnk.conf_int(alpha=0.05)
coefs_dnk, lo_dnk, hi_dnk, p_dnk = [], [], [], []

for w in all_weeks_dnk:
    col = str(w)
    if col in model_dnk.params.index:
        coefs_dnk.append(model_dnk.params[col])
        lo_dnk.append(ci_dnk.loc[col, 0])
        hi_dnk.append(ci_dnk.loc[col, 1])
        p_dnk.append(model_dnk.pvalues[col])
    else:
        coefs_dnk.append(0.0); lo_dnk.append(0.0); hi_dnk.append(0.0); p_dnk.append(0.0)

print(f"Denmark: N={int(model_dnk.nobs):,}  R2={model_dnk.rsquared:.4f}  DW={durbin
print("\nDENMARK POST-ISSUANCE DYNAMICS")
print("=" * 70)
print(f"{'Week':<8} {'Period':<10} {'Coef':>10} {'CI Lo':>10} {'CI Hi':>10} {'p':>8}
print("-" * 70)
for w, c, lo, hi, p in zip(all_weeks_dnk, coefs_dnk, lo_dnk, hi_dnk, p_dnk):
    period = "Pre" if w < -1 else "Ref" if w == -1 else "Post"
    ref    = "(ref)" if w == -1 else ("***" if p<0.01 else "*" if p<0.05 else "")
    print(f"w:<8} {period:<10} {c:>+10.4f} {lo:>10.4f} {hi:>10.4f} {p:>8.4f} {ref}")

```

Denmark: N=109 R2=0.0419 DW=2.7270

DENMARK POST-ISSUANCE DYNAMICS

Week	Period	Coef	CI Lo	CI Hi	p Sig
-10	Pre	+0.0000	0.0000	0.0000	1.0000
-9	Pre	+0.0000	0.0000	0.0000	1.0000
-8	Pre	+0.0000	0.0000	0.0000	1.0000
-7	Pre	+0.0000	0.0000	0.0000	1.0000
-6	Pre	+0.0000	0.0000	0.0000	1.0000
-5	Pre	+0.0000	0.0000	0.0000	1.0000
-4	Pre	+0.0000	0.0000	0.0000	1.0000
-3	Pre	+0.0000	0.0000	0.0000	1.0000
-2	Pre	+0.0000	0.0000	0.0000	1.0000
-1	Ref	+0.0000	0.0000	0.0000	1.0000 (ref)
0	Post	-0.0008	-0.0534	0.0519	0.9768
1	Post	-0.0081	-0.0510	0.0347	0.7104
2	Post	-0.0091	-0.0513	0.0331	0.6733
3	Post	-0.0099	-0.0536	0.0338	0.6567
4	Post	-0.0089	-0.0513	0.0335	0.6812
5	Post	-0.0079	-0.0523	0.0365	0.7275
6	Post	-0.0073	-0.0521	0.0375	0.7497
7	Post	-0.0077	-0.0525	0.0371	0.7359
8	Post	-0.0093	-0.0530	0.0344	0.6771
9	Post	-0.0109	-0.0538	0.0321	0.6200
10	Post	-0.0047	-0.0481	0.0387	0.8313
11	Post	-0.0095	-0.0520	0.0330	0.6626
12	Post	-0.0087	-0.0514	0.0340	0.6885
13	Post	-0.0091	-0.0520	0.0339	0.6788
14	Post	-0.0130	-0.0601	0.0342	0.5894
15	Post	-0.0084	-0.0508	0.0341	0.6988
16	Post	-0.0083	-0.0543	0.0377	0.7224
17	Post	-0.0116	-0.0565	0.0334	0.6143
18	Post	-0.0089	-0.0582	0.0405	0.7250
19	Post	-0.0086	-0.0520	0.0348	0.6963
20	Post	-0.0029	-0.0495	0.0437	0.9028

4.3 - Denmark chart

```

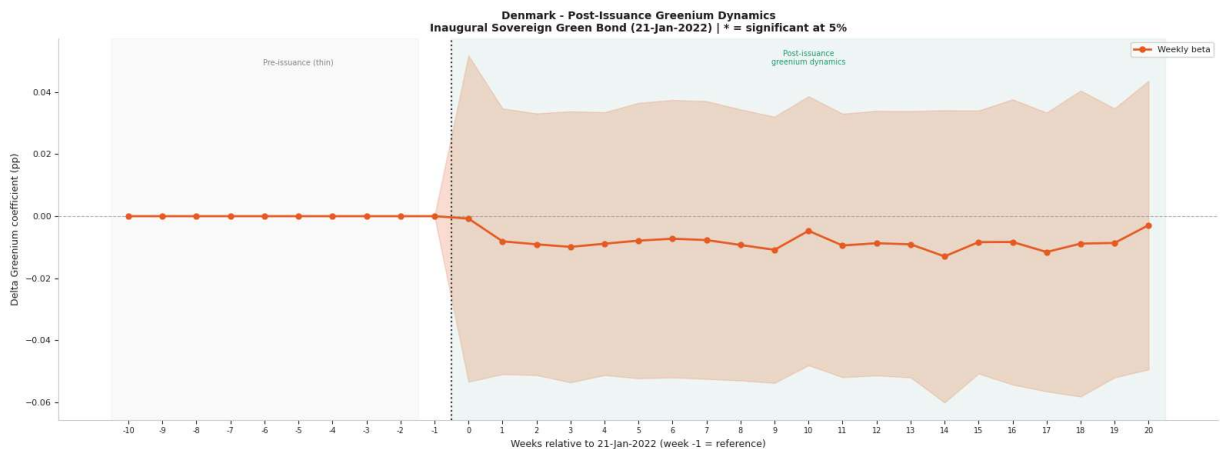
In [13]: fig, ax = plt.subplots(figsize=(16, 6))
fig.patch.set_facecolor('white'); style_ax(ax)
x_k=np.array(all_weeks_dnk); c_k=np.array(coefs_dnk); lo_k=np.array(lo_dnk); hi_k=n
ax.axvspan(x_k.min()-0.5,-1.5,alpha=0.03,color=GRAY)
ax.axvspan(-0.5,x_k.max()+0.5,alpha=0.06,color=TEAL)
ax.fill_between(x_k,lo_k,hi_k,alpha=0.2,color=ORANGE)
ax.plot(x_k,c_k,color=ORANGE,linewidth=2.0,marker="o",markersize=5,label="Weekly be
for w,c,p in zip(all_weeks_dnk,coefs_dnk,p_dnk):
    if p<0.05 and w!=-1: ax.plot(w,c,marker="*",color=BLACK,markersize=14,zorder=4)
ax.axhline(0,color=GRAY,linewidth=0.8,linestyle="--",alpha=0.7)
ax.axvline(-0.5,color=BLACK,linewidth=1.5,linestyle=":",alpha=0.9)
ylim=ax.get_ylim()
ax.text(-5,ylim[1]*0.85,"Pre-issuance (thin)",fontsize=7,color=GRAY,ha="center")
ax.text(10, ylim[1]*0.85, "Post-issuance\ngreenium dynamics",
        fontsize=7, color=TEAL, ha="center")

```

```

ax.set_xlabel("Weeks relative to 21-Jan-2022 (week -1 = reference)", fontsize=9)
ax.set_ylabel("Delta Greenium coefficient (pp)", fontsize=9)
ax.set_title("Denmark - Post-Issuance Greenium Dynamics\n"
             "Inaugural Sovereign Green Bond (21-Jan-2022) | * = significant at 5%"
             fontsize=10, fontweight="bold", color=BLACK)
ax.set_xticks(x_k); ax.set_xticklabels([str(w) for w in all_weeks_dnk], fontsize=7, c
ax.legend(fontsize=8, facecolor='white', edgecolor="#cccccc", labelcolor=BLACK)
plt.tight_layout()
out=os.path.join(OUTPUT_FOLDER, "event_study_denmark.png")
plt.savefig(out, dpi=150, bbox_inches="tight", facecolor='white'); plt.show()
print(f"Saved: {out}")

```



Saved: ..\output\regression_europe\event_study_denmark.png

Part 5 - Pooled Event Study (Germany + Denmark)

Limitation note

Pooling Germany (treatment Sep 2020) and Denmark (treatment Jan 2022) in one regression imposes the assumption that the greenium dynamic is identical across both events. These events occurred 16 months apart in very different macro environments (post-COVID recovery vs pre-rate-hike). The pooled chart is useful for a thesis overview figure but the country-specific results in Parts 3 and 4 are primary.

5.1 - Pooled event time assignment

```

In [14]: panel_es = panel_fd.copy()
panel_es["event_time"] = np.nan

for country, treat_date in TREATMENT_DATES.items():
    mask = panel_es["country"] == country
    country_dates = np.sort(panel_es.loc[mask, "date"].unique())
    treat_idx = np.searchsorted(country_dates, np.datetime64(treat_date))

```

```

date_to_event = {d: int(i - treat_idx) for i, d in enumerate(country_dates)}
panel_es.loc[mask, "event_time"] = panel_es.loc[mask, "date"].map(date_to_event)

panel_es["event_week"] = (panel_es["event_time"] / 5).apply(np.floor).astype(int)
WEEK_MIN, WEEK_MAX = -10, 20
panel_es = panel_es[panel_es["event_week"].between(WEEK_MIN, WEEK_MAX)].copy()

print(f"Pooled: {len(panel_es):,} obs | weeks {WEEK_MIN} to {WEEK_MAX}")
print(panel_es.groupby("country")["event_week"].describe()[["count", "min", "max"]].t

```

Pooled: 214 obs | weeks -10 to 20

	count	min	max
country			
Denmark	109.0	-1.0	20.0
Germany	105.0	0.0	20.0

5.2 - Pooled regression

```

In [15]: event_dummies = pd.get_dummies(panel_es["event_week"].astype(str),
                                         drop_first=False, dtype=float)
if "-1" in event_dummies.columns:
    event_dummies = event_dummies.drop(columns=["-1"])

panel_es_reg = pd.concat([panel_es, event_dummies], axis=1)
event_cols = list(event_dummies.columns)

pair_dummies_es = pd.get_dummies(panel_es_reg["pair_id"], drop_first=True, dtype=float)
panel_es_reg = pd.concat([panel_es_reg, pair_dummies_es], axis=1)
fe_cols_es = list(pair_dummies_es.columns)

X_es = sm.add_constant(panel_es_reg[event_cols + fe_cols_es])
model_es = OLS(panel_es_reg["greenium_fd"], X_es).fit(cov_type="HC3")

all_weeks = sorted(range(WEEK_MIN, WEEK_MAX + 1))
ci_es = model_es.conf_int(alpha=0.05)
coefs, ci_lo, ci_hi, pvals = [], [], [], []

for w in all_weeks:
    col = str(w)
    if col in model_es.params.index:
        coefs.append(model_es.params[col])
        ci_lo.append(ci_es.loc[col, 0])
        ci_hi.append(ci_es.loc[col, 1])
        pvals.append(model_es.pvalues[col])
    else:
        coefs.append(0.0); ci_lo.append(0.0); ci_hi.append(0.0); pvals.append(1.0)

print(f"Pooled: N={int(model_es.nobs):,} R2={model_es.rsquared:.4f}")
print("\nPPOOLED COEFFICIENTS")
print("=" * 60)
for w, c, lo, hi, p in zip(all_weeks, coefs, ci_lo, ci_hi, pvals):
    period = "Pre " if w < -1 else "Ref " if w == -1 else "Post"
    ref = "(ref)" if w == -1 else ("****" if p < 0.01 else "***" if p < 0.05 else "**")
    print(f"{w:<8} {period:<8} {c:>+10.5f} {lo:>10.5f} {hi:>10.5f} {ref:>6}")

```

Pooled: N=214 R2=0.0279

POOLED COEFFICIENTS

=====				
-10	Pre	+0.00000	0.00000	0.00000
-9	Pre	+0.00000	0.00000	0.00000
-8	Pre	+0.00000	0.00000	0.00000
-7	Pre	+0.00000	0.00000	0.00000
-6	Pre	+0.00000	0.00000	0.00000
-5	Pre	+0.00000	0.00000	0.00000
-4	Pre	+0.00000	0.00000	0.00000
-3	Pre	+0.00000	0.00000	0.00000
-2	Pre	+0.00000	0.00000	0.00000
-1	Ref	+0.00000	0.00000	0.00000 (ref)
0	Post	-0.00556	-0.05043	0.03930
1	Post	-0.00838	-0.05080	0.03403
2	Post	-0.00896	-0.05120	0.03328
3	Post	-0.00876	-0.05133	0.03382
4	Post	-0.00906	-0.05135	0.03323
5	Post	-0.00767	-0.05039	0.03506
6	Post	-0.00765	-0.05055	0.03525
7	Post	-0.00838	-0.05119	0.03444
8	Post	-0.00875	-0.05133	0.03383
9	Post	-0.00788	-0.05038	0.03461
10	Post	-0.00667	-0.04919	0.03585
11	Post	-0.00884	-0.05114	0.03346
12	Post	-0.00838	-0.05072	0.03396
13	Post	-0.00864	-0.05104	0.03376
14	Post	-0.01110	-0.05459	0.03238
15	Post	-0.00770	-0.04998	0.03458
16	Post	-0.00828	-0.05141	0.03484
17	Post	-0.01049	-0.05347	0.03248
18	Post	-0.00844	-0.05233	0.03546
19	Post	-0.00843	-0.05093	0.03407
20	Post	-0.00547	-0.04875	0.03781

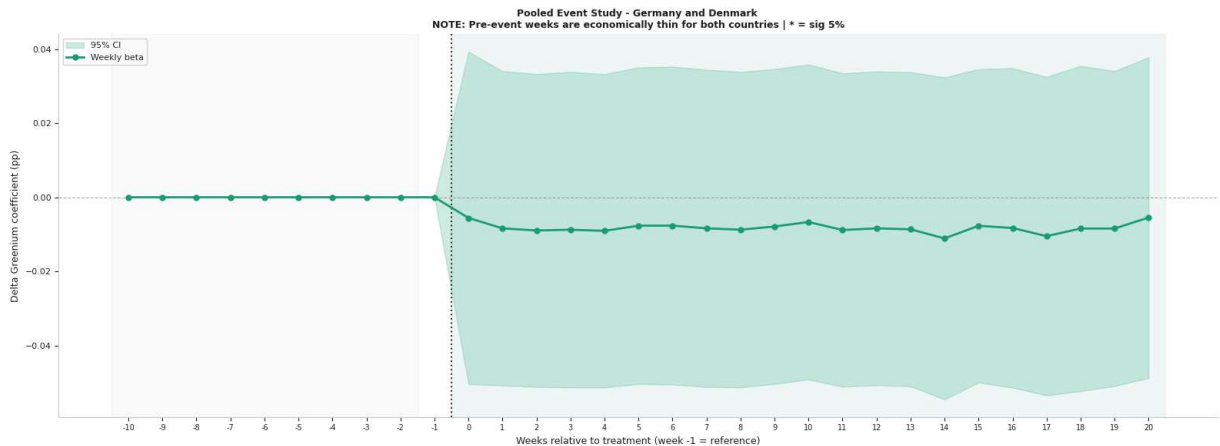
5.3 - Pooled chart

```
In [16]: fig, ax = plt.subplots(figsize=(16, 6))
fig.patch.set_facecolor('white'); style_ax(ax)
x=np.array(all_weeks); c=np.array(coefs); lo=np.array(ci_lo); hi=np.array(ci_hi)
ax.axvspan(x.min()-0.5,-1.5,alpha=0.03,color=GRAY)
ax.axvspan(-0.5,x.max()+0.5,alpha=0.06,color=TEAL)
ax.fill_between(x,lo,hi,alpha=0.2,color=TEAL,label="95% CI")
ax.plot(x,c,color=TEAL,linewidth=2.0,marker="o",markersize=5,label="Weekly beta",z
for w,c,p in zip(all_weeks,coefs,pvals):
    if p<0.05 and w!=-1: ax.plot(w,c,marker="*",color=ORANGE,markersize=14,zorder=4
ax.axhline(0,color=GRAY,linewidth=0.8,linestyle="--",alpha=0.7)
ax.axvline(-0.5,color=BLACK,linewidth=1.5,linestyle=":",alpha=0.9)
ax.set_xlabel("Weeks relative to treatment (week -1 = reference)",fontsize=9)
ax.set_ylabel("Delta Greenium coefficient (pp)",fontsize=9)
ax.set_title("Pooled Event Study - Germany and Denmark \nNOTE: Pre-event weeks are
ax.set_xticks(x); ax.set_xticklabels([str(w) for w in all_weeks],fontsize=7,color=B
ax.legend(fontsize=8,facecolor='white',edgecolor="#cccccc",labelcolor=BLACK)
plt.tight_layout()
```

```

out=os.path.join(OUTPUT_FOLDER,"event_study_pooled_europe.png")
plt.savefig(out,dpi=150,bbox_inches="tight",facecolor='white'); plt.show()
print(f"Saved: {out}")

```



Saved: ..\output\regression_europe\event_study_pooled_europe.png

Part 6 - Side-by-Side Comparison

```

In [17]: fig, axes = plt.subplots(1, 2, figsize=(20, 6))
fig.patch.set_facecolor('white')

specs = [
    ("Germany", BLUE, all_weeks_deu, coefs_deu, lo_deu, hi_deu, p_deu,
     "Inaugural Green Bund | 2-Sep-2020 \nPost-issuance price discovery"),
    ("Denmark", ORANGE, all_weeks_dnk, coefs_dnk, lo_dnk, hi_dnk, p_dnk,
     "Inaugural Green Bond | 21-Jan-2022 \nPost-issuance greenium dynamics"),
]

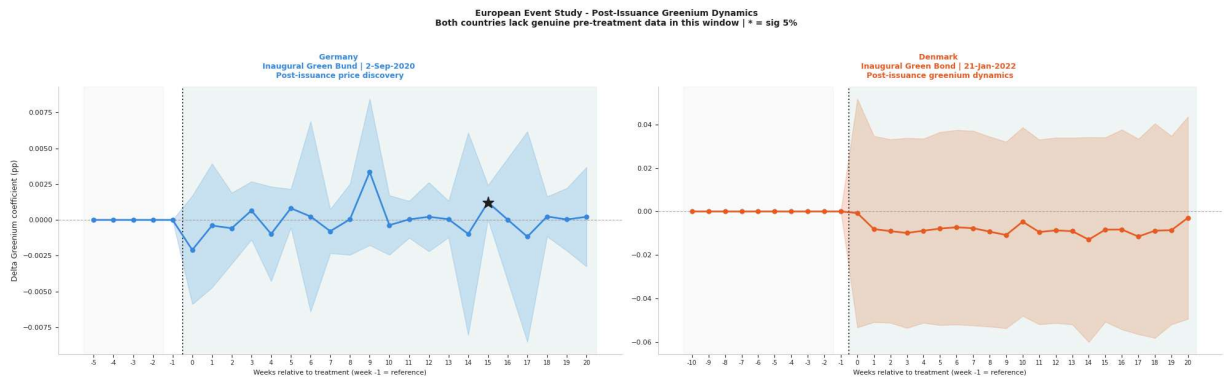
for ax, (country, color, weeks, coefs_, lo_, hi_, p_, subtitle) in zip(axes, specs):
    style_ax(ax)
    x=np.array(weeks); c=np.array(coefs_); l=np.array(lo_); h=np.array(hi_)
    ax.axvspan(x_.min()-0.5,-1.5,alpha=0.03,color=GRAY)
    ax.axvspan(-0.5,x_.max()+0.5,alpha=0.06,color=TEAL)
    ax.fill_between(x_,l_,h_,alpha=0.2,color=color)
    ax.plot(x_,c_,color=color,linewidth=2.0,marker="o",markersize=5,zorder=3)
    for w,c,p in zip(weeks,coefs_,p_):
        if p<0.05 and w!=-1: ax.plot(w,c,marker="*",color=BLACK,markersize=14,zorde
    ax.axhline(0,color=GRAY,linewidth=0.8,linestyle="--",alpha=0.7)
    ax.axvline(-0.5,color=BLACK,linewidth=1.5,linestyle=":",alpha=0.9)
    ax.set_xticks(x_); ax.set_xticklabels([str(w) for w in weeks],fontsize=7,color=
    ax.set_xlabel("Weeks relative to treatment (week -1 = reference)",fontsize=8)
    ax.set_title(f"{country} \n{subtitle}",fontsize=9,fontweight="bold",color=color

axes[0].set_ylabel("Delta Greenium coefficient (pp)",fontsize=9)
fig.suptitle("European Event Study - Post-Issuance Greenium Dynamics\n"
             "Both countries lack genuine pre-treatment data in this window | * = s
             fontsize=10, fontweight="bold", color=BLACK, y=1.02)
plt.tight_layout()
out=os.path.join(OUTPUT_FOLDER,"event_study_comparison_europe.png")

```



```
plt.savefig(out,dpi=150,bbox_inches="tight",facecolor='white'); plt.show()
print(f"Saved: {out}")
```



Saved: ..\output\regression_europe\event_study_comparison_europe.png

Part 7 - Joint F-Test on Post-Event Dummies

What we test here

Because both countries lack pre-treatment data, we cannot run a parallel trends F-test. Instead we run a joint F-test on post-event dummies to assess whether the post-issuance greenium dynamics are statistically structured (non-random).

H0: All post-event week coefficients = 0 (no systematic post-issuance pattern)

A significant result means the greenium showed a structured dynamic after inaugural issuance. This complements the regression notebook M5 Post coefficient by showing the SHAPE of the post-event response, not just the level shift.

```
In [18]: def post_event_joint_test(model, all_weeks, label):
    post_cols = [str(w) for w in all_weeks if w > 0 and str(w) in model.params.index]
    if not post_cols:
        print(f"[{label}] No post-event dummies.")
        return
    wald = model.wald_test(post_cols, use_f=True)
    f_stat = float(wald.statistic)
    p_val = float(wald.pvalue)
    result = "SIGNIFICANT: structured post-issuance dynamics" if p_val < 0.1 else
    print(f"\n[{label}]")
    print(f"  Weeks tested: {len(post_cols)} post-event dummies")
    print(f"  Joint F({len(post_cols)}, {int(model.df_resid)}) = {f_stat:.4f}  p = ")
    print(f"  Result: {result}")

    print("JOINT F-TEST ON POST-EVENT DUMMIES - EUROPE")
    print("=" * 65)
    print("H0: All post-event coefficients = 0")
```

```

print("NOTE: This tests post-issuance dynamics, NOT parallel trends")
print()
post_event_joint_test(model_es, all_weeks, "POOLED")
post_event_joint_test(model_deu, all_weeks_deu, "GERMANY")
post_event_joint_test(model_dnk, all_weeks_dnk, "DENMARK")
print()
print("INTERPRETATION:")
print(" Significant -> greenium had structured dynamics after inaugural issuance")
print(" Not significant -> post-event coefficients jointly indistinguishable from")
print(" Consistent with M5 Post coefficient from regression notebook?")
print(f" Check regression notebook M5_Germany and M5_Denmark for comparison")

```

JOINT F-TEST ON POST-EVENT DUMMIES - EUROPE

=====

H0: All post-event coefficients = 0

NOTE: This tests post-issuance dynamics, NOT parallel trends

[POOLED]

Weeks tested: 20 post-event dummies

Joint F(20, 191) = 0.1596 p = 1.0000

Result: NOT significant: no systematic post-issuance pattern

[GERMANY]

Weeks tested: 20 post-event dummies

Joint F(20, 84) = 0.6030 p = 0.9001

Result: NOT significant: no systematic post-issuance pattern

[DENMARK]

Weeks tested: 20 post-event dummies

Joint F(20, 87) = 0.0996 p = 1.0000

Result: NOT significant: no systematic post-issuance pattern

INTERPRETATION:

Significant -> greenium had structured dynamics after inaugural issuance

Not significant -> post-event coefficients jointly indistinguishable from zero

Consistent with M5 Post coefficient from regression notebook?

Check regression notebook M5_Germany and M5_Denmark for comparison

Part 8 - Summary

```

In [19]: print("=" * 65)
print("EUROPEAN EVENT STUDY - SUMMARY")
print("=" * 65)

for label, weeks, coefs_, pvals_ in [
    ("POOLED", all_weeks, coefs, pvals),
    ("GERMANY", all_weeks_deu, coefs_deu, p_deu),
    ("DENMARK", all_weeks_dnk, coefs_dnk, p_dnk),
]:
    post_coefs = [c for w, c in zip(weeks, coefs_) if w > 0]
    post_sig5 = sum(1 for w, p in zip(weeks, pvals_) if w > 0 and p < 0.05)

```



```

post_sig10 = sum(1 for w,p in zip(weeks,pvals_) if w > 0 and p < 0.10)
mean_post = sum(post_coefs)/len(post_coefs) if post_coefs else 0
verdict = "Structured compression" if post_sig5 >= 2 else "Weak p
print(f" \n[{label}])")
print(f" Mean post-event beta: {mean_post:+.5f} pp")
print(f" Significant post weeks (p<0.05): {post_sig5}/{len(post_coefs)}")
print(f" Significant post weeks (p<0.10): {post_sig10}/{len(post_coefs)}")
print(f" Verdict: {verdict}")

print()
print("=" * 65)
print("KEY THESIS IMPLICATIONS")
print("=" * 65)
print()
print("1. PARALLEL TRENDS:")
print(" Cannot be tested from event study for Germany or Denmark.")
print(" Both issued green bonds AT the treatment date.")
print(" Validation comes from stacked DiD Stack A (regression notebook).")
print()
print("2. POST-ISSUANCE DYNAMICS:")
print(" See coefficient tables and charts above.")
print(" Negative significant post-event betas = greenium compression after issuan
print()
print("3. COMPARISON WITH ASIA:")
print(" Asia: null H3, no post-event dynamics.")
print(" Europe: check results above.")
print(" If Europe shows systematic compression, this supports market maturity int
print()
print("Output files:")
for f in ["event_study_germany.png", "event_study_denmark.png",
         "event_study_pooled_europe.png", "event_study_comparison_europe.png"]:
    print(f" {os.path.join(OUTPUT_FOLDER, f)}")

```

=====

EUROPEAN EVENT STUDY - SUMMARY

=====

[POOLED]

Mean post-event beta: -0.00840 pp
 Significant post weeks ($p < 0.05$): 0/20
 Significant post weeks ($p < 0.10$): 0/20
 Verdict: No pattern

[GERMANY]

Mean post-event beta: +0.00009 pp
 Significant post weeks ($p < 0.05$): 1/20
 Significant post weeks ($p < 0.10$): 1/20
 Verdict: Weak pattern

[DENMARK]

Mean post-event beta: -0.00864 pp
 Significant post weeks ($p < 0.05$): 0/20
 Significant post weeks ($p < 0.10$): 0/20
 Verdict: No pattern

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KEY THESIS IMPLICATIONS

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1. PARALLEL TRENDS:

Cannot be tested from event study for Germany or Denmark.
 Both issued green bonds AT the treatment date.
 Validation comes from stacked DiD Stack A (regression notebook).

2. POST-ISSUANCE DYNAMICS:

See coefficient tables and charts above.
 Negative significant post-event betas = greenium compression after issuance.

3. COMPARISON WITH ASIA:

Asia: null H3, no post-event dynamics.
 Europe: check results above.

If Europe shows systematic compression, this supports market maturity interpretation.

Output files:

..\output\regression_europe\event_study_germany.png
 ..\output\regression_europe\event_study_denmark.png
 ..\output\regression_europe\event_study_pooled_europe.png
 ..\output\regression_europe\event_study_comparison_europe.png

Appendix - Comparison with Asia event study

Dimension	India	Indonesia	Germany	Denmark
Pre-treatment data	YES ~120 days	Very limited ~55 days	NO	NO

Dimension	India	Indonesia	Germany	Denmark
Parallel trends testable here	YES	Weakly	NO	NO
Where PT is validated	This event study	Stacked DiD	Twin-bond design	Stacked DiD Stack A
Post-issuance dynamics	Null	Null	Check results	Check results

The European event study documents post-issuance dynamics rather than providing a parallel trends validation. The DiD identification for Europe relies on the twin-bond design and stacked DiD specification, not on this event study.

References

- MacKinlay, A.C. (1997). Event studies in economics and finance. JEL 35(1), 13-39.
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- Ando, S. et al. (2023). European sovereign green bonds. IMF Working Paper.
- Zerbib, O.D. (2019). Green bond premium. JBF 98, 39-60.